

Exact Bit-Flip Noise and Walsh Spectrum of a Frozen Hénon Core

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Abstract

We compute the exact Boolean spectrum of the full-core generation indicator on a frozen sixteen-label support. The indicator is one precisely when the pivot S_9 is present and at least two of four direction blocks are hit. Its truth table has 30400 ones. An integer Walsh–Hadamard transform has 1024 nonzero coefficients and maximum degree 10; a second transform gives the complete ordered-pair Hamming autocorrelation and hence the exact bit-flip noise polynomial. Independent mask-level checks, a ten-variable symbolic calculation, clean replay, and thirteen hostile mutations certify the receipt. This is a finite Boolean result, with scope firewall `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Predicate and transform

For $A \subseteq L$, define

$$F(A) = \mathbf{1}\{S_9 \in A\} \mathbf{1}\{A \text{ meets at least two of } B_1, B_2, B_3, B_4\},$$

where

$$B_1 = \{S_1\}, \quad B_2 = \{S_{16}\}, \quad B_3 = \{S_7, S_{15}\}, \quad B_4 = \{S_3, S_4, S_8, S_{11}, S_{12}\}.$$

The six other labels do not affect F . We enumerate all 65536 masks and use the convention

$$W(S) = \sum_{A \subseteq L} F(A) (-1)^{|A \cap S|}.$$

Proposition 1. *The Walsh table has 1024 nonzero entries, degree at most 10, and satisfies*

$$\sum_S W(S)^2 = 2^{16} 30400.$$

Certificate argument. The six dummy variables force every coefficient containing a dummy bit to vanish. The remaining ten-variable multilinear polynomial is a pivot times an inclusion–exclusion expression for at least two block hits. Expanding it over signs gives the 1024 active coefficients; the producer and independent symbolic script agree coefficient by coefficient. Parseval gives the stated energy identity. $\square$

2 Bit-flip autocorrelation

For Hamming distance h , let

$$C_h = \sum_{d_H(A,B)=h} F(A)F(B).$$

The exact vector for $h = 0, \dots, 16$ is

$$(C_0, \dots, C_{16}) = (30400, 445696, 3068864, 13137152, 39054016, 85341312, \\ 141543616, 181366144, 180954240, 140539776, 84251904, 38278400, \\ 12756480, 2943488, 420480, 28032, 0).$$

It sums to 30400^2 . With independent bit flips of rate ε , the normalized agreement probability is equivalently

$$2^{-16} \sum_h C_h \varepsilon^h (1 - \varepsilon)^{16-h} = 2^{-32} \sum_S W(S)^2 (1 - 2\varepsilon)^{|S|}.$$

3 Audit and limits

The receipt binds C73/C76/C78 evidence and manifests by SHA-256. Its canonical evidence hash is the concatenation of

6fc49cad02956f463b1e37d017506f43
7edce6717414da74770ad94913ccefa1.

The independent checker re-evaluates the generation predicate and the C78 distance-zero boundary for every mask. The symbolic check uses the explicit active coordinate order $(S_1, S_3, S_4, S_7, S_8, S_9, S_{11}, S_{12}, S_{15}, S_{16})$ and integrates out the six dummies as a factor 64. No arithmetic/local, Euler-factor, root-number, automorphy, Burnside-ring/table-of-marks, or Hilbert–Polya statement is made.